



Welcome to Section

Week 6

Introduction to Data Structures



Slides by Cameron Mohne and Maggie Lee





01

Check-in

How are you doing?

02

Recap

Going over lists and dictionaries again



Agenda

03

Index Game

Refreshing list indexing!



04

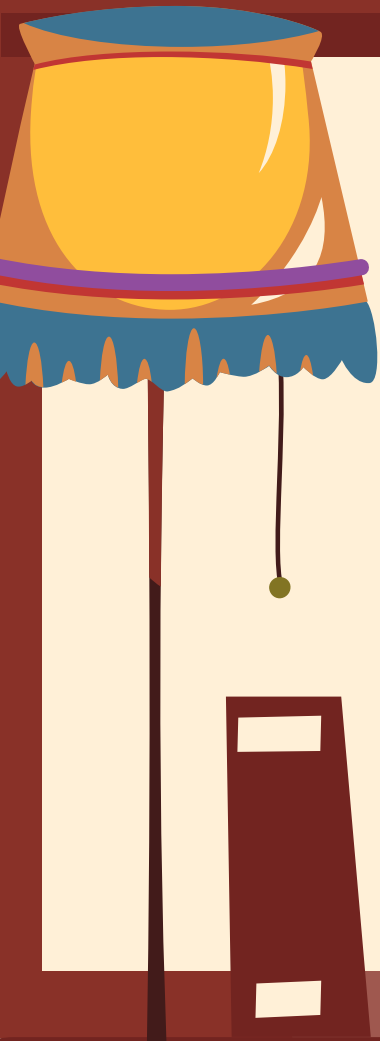
List Practice

Get practice creating and modifying lists!

05

Heads Up

Get practice functionally utilizing lists!



Check-in



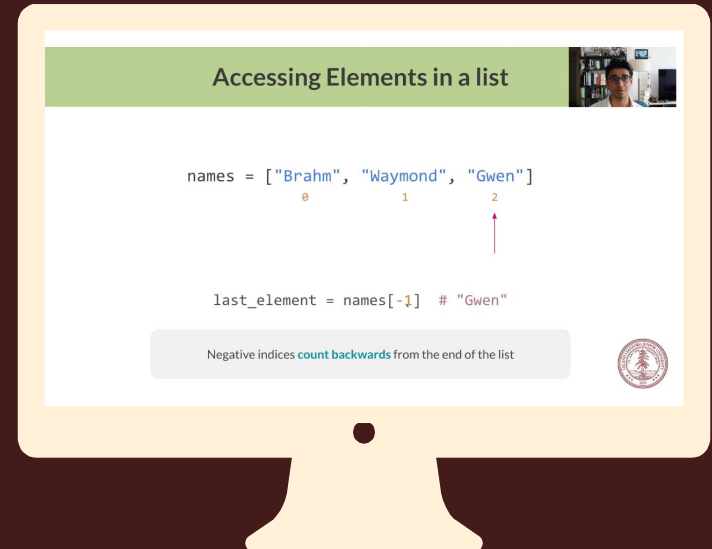
Unfortunately, it is the end of Code in Place.
Let's reminisce a bit, shall we? What was your
favorite problem, section, or memory from
your time here? I'll go first!



A stylized illustration of a stage. On the left, a yellow spotlight on a green stand shines a beam of light across the stage. The stage floor is dark brown, and the background is a lighter brown. Two books are floating in the air: a red one on the left and a purple one on the right. The word "Questions?" is written in a large, white, serif font at the bottom center of the stage.

Questions?

Recap



Data Structures



List

An ordered collection of values. Usually used to store a "list" of similar data. One example might be an attendance roster. It's usually in order alphabetically (but it doesn't have to be) and stores a list of names!



Dictionary

An unordered collection of key/value pairs. Usually used to store relational data. One example is a phone book or a contact list in a phone. Each name has a number associated with it!





Lists



Some useful documentation:

- `list_name = []`
- `list_name = [value_1, value_2, ..., value_n]`
- `list_name.append(elem)`
- `list_name.insert(index, elem)`
- `list_name.remove(elem)`
- `popped_item = list_name.pop(index)`
- `accessed_item = list_name[index]`



Lists

Some basic reminders regarding lists:

- **Lists are ordered**, meaning you can access items (often called elements) using their *index* (what spot they are) in the list.
 - Indexing starts at 0 in Python! This means that the very first element in a list has index 0.
- **Lists are mutable**, if you make changes to a list in a helper function, they actually modify the list!
- **Lists don't need to contain data of the same type**. It is just more common.
 - This means that a singular list can contain strings, integers, booleans, floats, and more.
- We can find the length of a list using `len(list_name)`





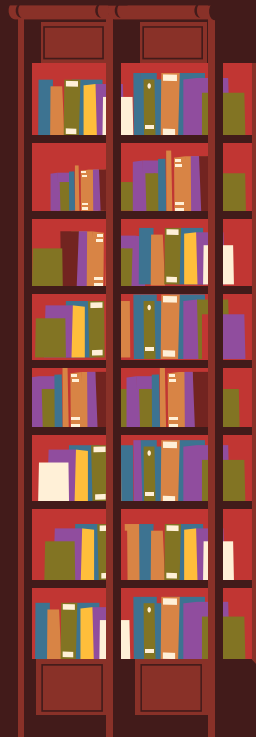
Dictionaries



Some useful documentation:

- `dict_name = {}`
- `dict_name = {key: value, key2: value_2, ...}`
- `dict_name[key] = value`
- `del dict_name[key]`
- `value = dict_name.pop(key, None)`
- `value = dict_name[key]`
- `value = dict_name.get(key)`

Maggie's favorite keyword!





Dictionaries



Some basic reminders regarding dictionaries:

- Often short handed in writing/speech to dict.
- **Also mutable**, just like lists
- **Make sure keys are in a dictionary** or have some check to see if they are there before trying to access or delete data.
 - This is usually needed when trying to use user input in conjunction with dictionaries
- **Keys have to be unique**
- **Keys can be anything that is *hashable***, are often strings, though.





Questions?

Index Game



List Practice





Questions?

Heads Up



Problem Premise



Heads Up is a game where one person takes the role of a "guesser".

A word will be generated at random, and everyone who isn't the guesser will provide hints as to what the word is, without saying the actual word.

The guessers job is to guess what the word is as fast as possible! An example of a round of this game is shown on the right.





Problem Overview

We are all going to become game developers today, as our job is to recreate the functionality of this game!

We are going to take a list of words from a file and then use that list with our knowledge of Python to recreate a console-based version of this game!



End Goal

Example Input



(Example Word List)

Apple
Banana
Pear
Blueberry
Strawberry
Longan
Lychee
Tamarind
Orange



Example Output

The word to guess: Lychee
Hit 'enter' to get the next word.

The word to guess: Pear
Hit 'enter' to get the next word.

The word to guess: Longan
Hit 'enter' to get the next word.

...





Problem Details

Words

We provide a list of words for you to use in a file named `"cswords.txt"`.

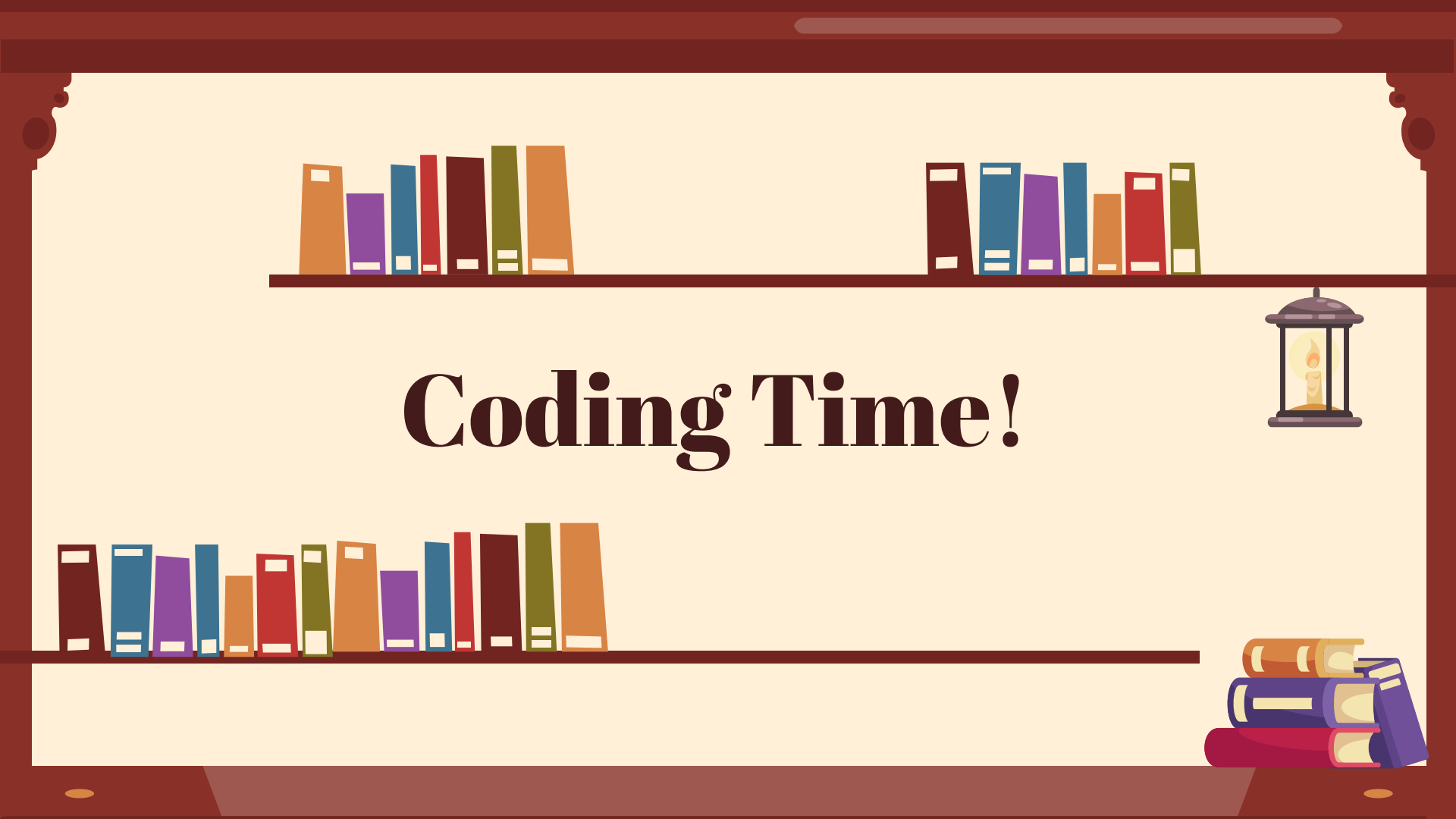
Files

We are given a function that will read words from a file for us called `get_words_from_file(...)`.

Game Loop

We want to repeat the game infinitely (or until we hit the `stop` button at the top of the IDE)





Coding Time!

Extensions

Organization:

Cycle through names so people have turns. Great for virtual play!

```
% python main.py
It is now Nick's turn. Press enter to get the word.
Nick's word is Variable
Hit enter to move on to the next person.

It is now Maggie's turn. Press enter to get the word.
Maggie's word is If Statement
Hit enter to move on to the next person.

It is now Yuliia's turn. Press enter to get the word.
Yuliia's word is Toaster
Hit enter to move on to the next person.

It is now Nick's turn. Press enter to get the word.
```

Categories:

Make your own word lists and add categories to play with!
(Bonus points for using a dictionary!)

```
% python main.py
Pick a category from the following:
  cip
  fruit
  movies
  brands
Which category would you like?: fruit

It is now Nick's turn. Press enter to get the word.
Nick's word is apple
Hit enter to move on to the next person.

It is now Maggie's turn. Press enter to get the word.
Maggie's word is apple
Hit enter to move on to the next person.

It is now Yuliia's turn. Press enter to get the word.
Yuliia's word is durian
Hit enter to move on to the next person.
```



Thank You

This experience wouldn't be possible without all you amazing students! It was a joy working with you during this iteration of Code in Place!!!

